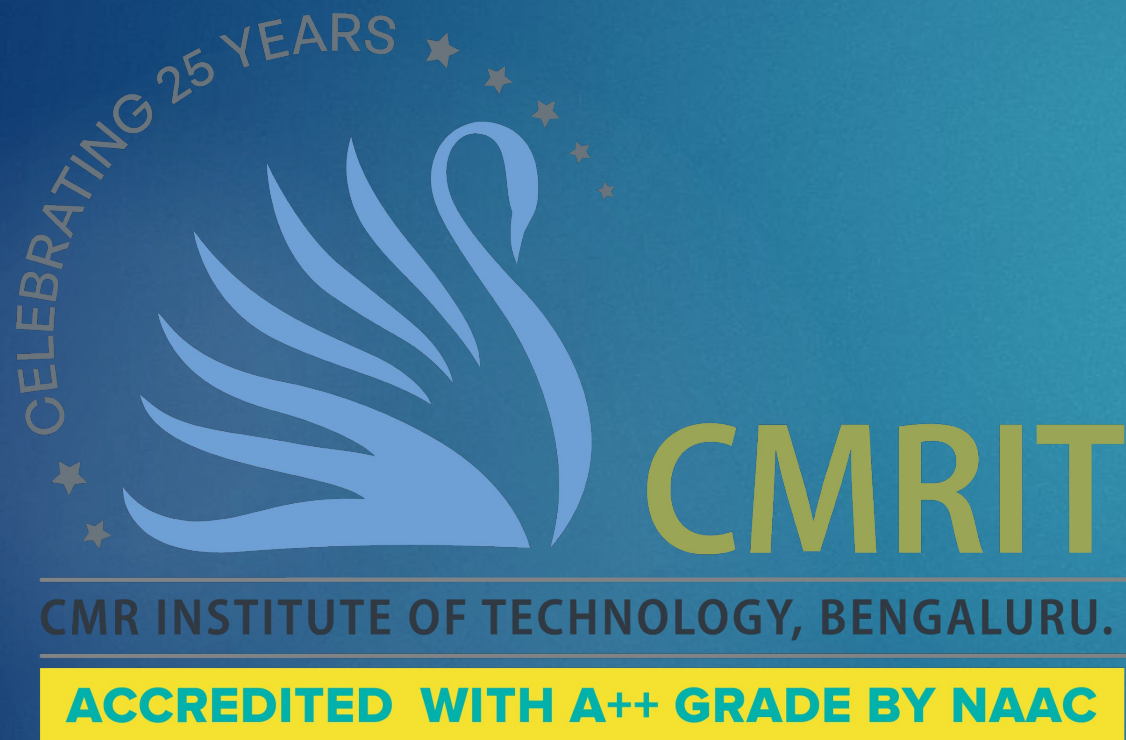


PASSENGER INFORMATION SYSTEM



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Sir.....

INTRODUCTION

- Passenger Information System (PIS) is used to provide real-time updates about buses
- It helps passengers know bus location, routes, and arrival time
- Traditional systems do not provide accurate timing
- GPS-based systems improve accuracy and reliability
- It is an important part of smart transportation systems

PROBLEM STATEMENT

- Passengers face uncertainty in bus arrival time
- Long waiting time at bus stops
- Inefficient public transport system
- Need for a system that provides live tracking and updates

OBJECTIVES

- To design a system for real-time bus tracking using GPS
- To provide accurate arrival information
- To reduce passenger waiting time
- To improve public transport efficiency
- To develop a low-cost and reliable solution

PROPOSED SOLUTION

- Real-Time Tracking:

Uses GPS module to capture live bus location.

- Processing Unit:

ESP32/Arduino microcontroller processes GPS data and controls the system.

- Data Transmission:

Location data is sent to cloud/server using Wi-Fi or GSM.

- Web Application:

Displays real-time bus location, route details, and updates to users.

- User Interface:

Passengers can view bus location and track movement easily.

- Local Display:

LCD/Display shows basic information like location and system status.

- Cost-Effective Design:

Built using ESP32, GPS module, jumper wires, and simple components.

- Scalable System:

Can be extended for multiple buses and different routes.

HARDWARE COMPONENTS

NEO-6M GPS Module

Receives signals from satellites to determine location.
Provides real-time latitude and longitude coordinates.

ESP32

Acts as the main controller of the system.
Processes GPS data and sends it via WiFi.

LCD Display

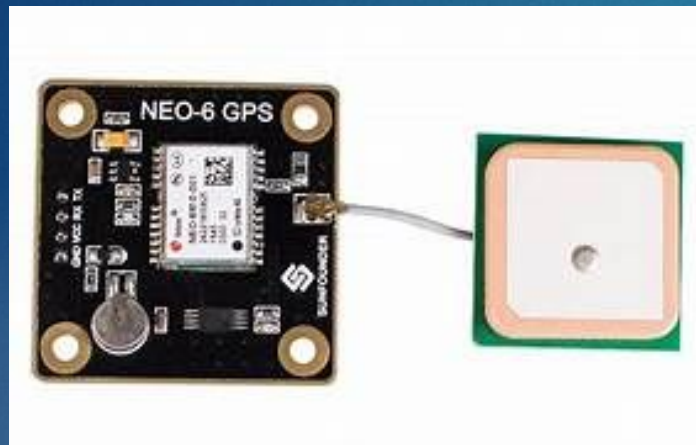
Displays location and system information.
Provides output to the user in real time.

Breadboard & Wires

Connects all components without soldering.
Helps in easy circuit setup and testing.

HARDWARE COMPONENTS

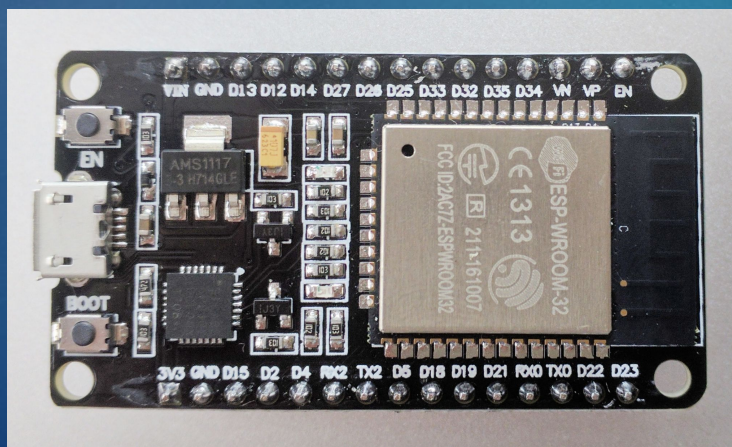
NEO-6M GPS Module



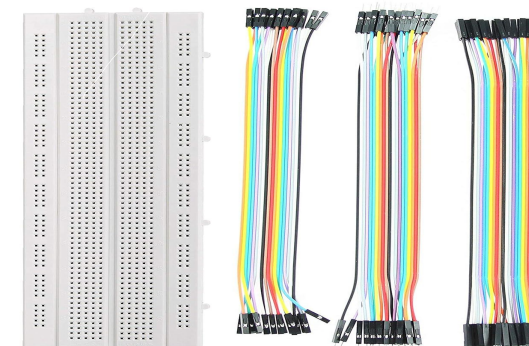
LCD Display



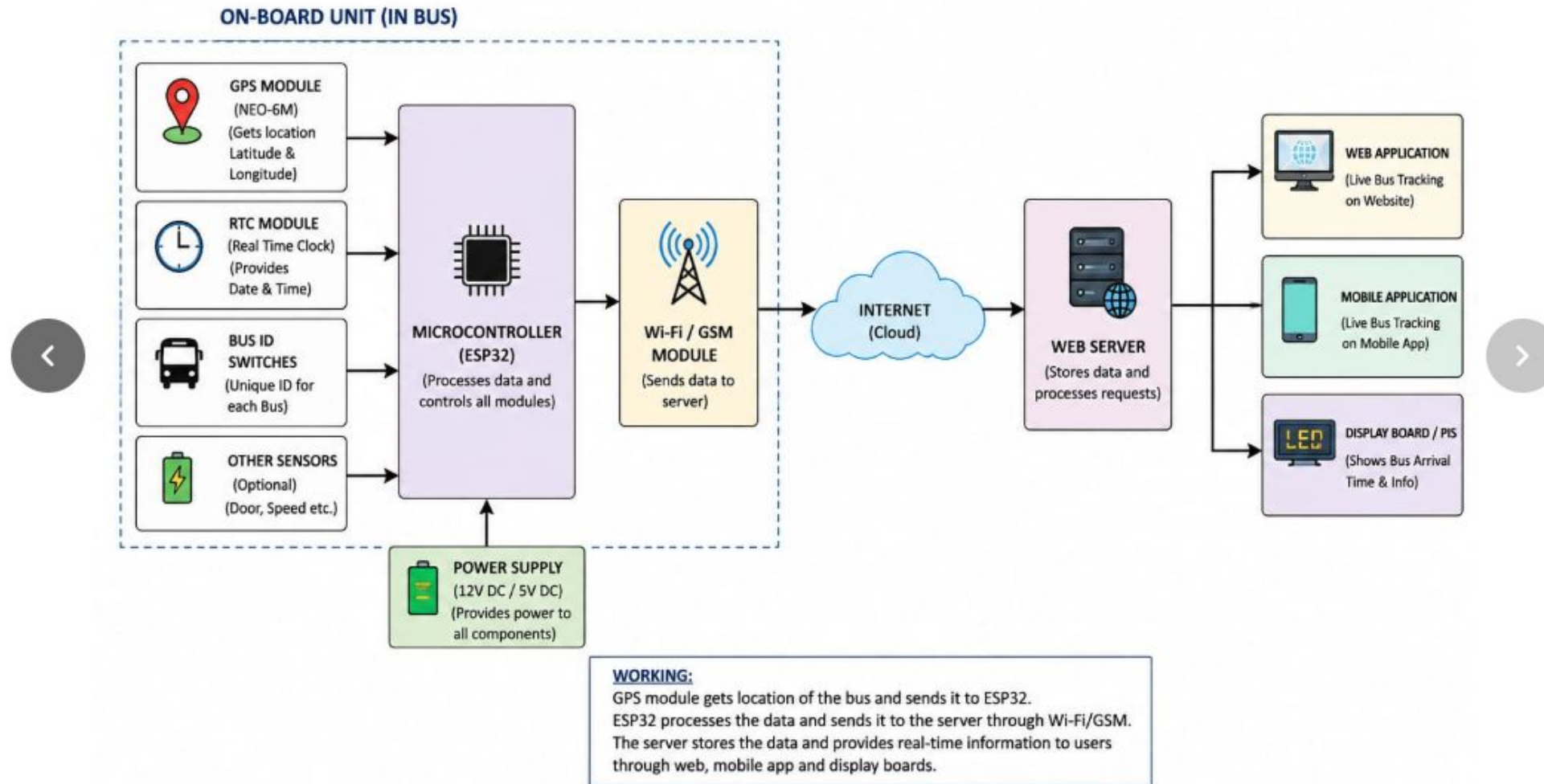
ESP32



Breadboard & Wires

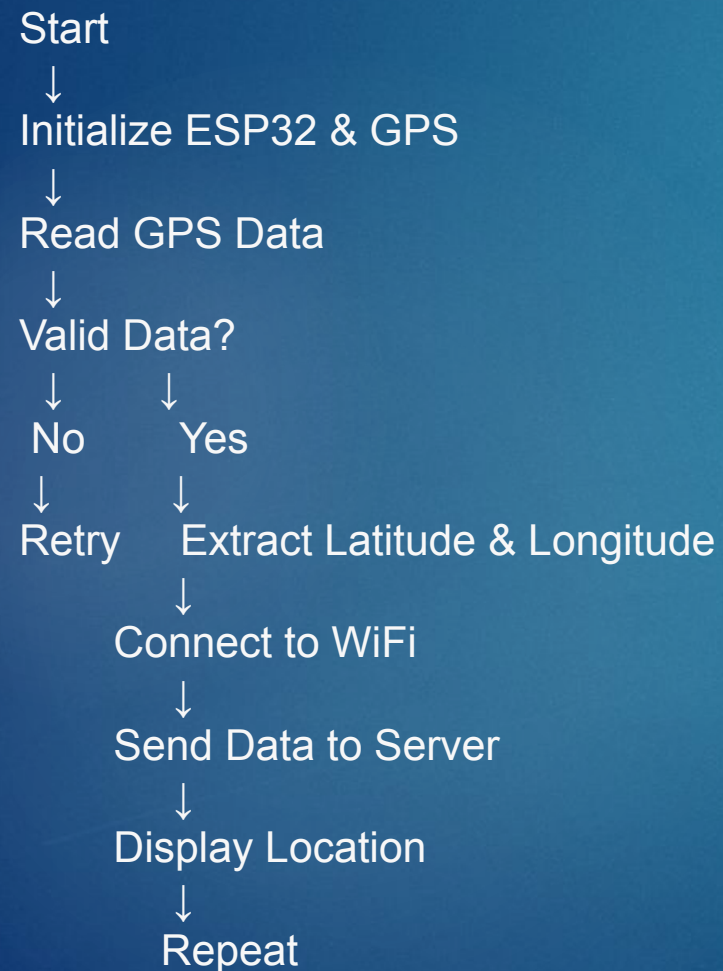


BLOCK DIAGRAM



FLOWCHART AND ALGORITHM

FLOWCHART



ALGORITHM

- Start the system
- Initialize ESP32 and GPS module
- Establish serial communication with GPS
- Read GPS data continuously
- Check if data is valid
- Extract latitude and longitude
- Connect ESP32 to WiFi network
- Send location data to server/cloud
- Display location on LCD/mobile app
- Repeat the process

ADVANTAGES & APPLICATIONS

❖ Advantages:

- Real-time tracking
- Reduces waiting time
- Easy to use
- Improves transport efficiency

❖ Applications:

- City bus systems
- School/college buses
- Railway tracking
- Logistics and fleet management

CONCLUSION AND FUTURE SCOPE

❖ **Conclusion:**

- Provides accurate and real-time passenger information
- Enhances convenience and reliability
- Supports smart city development

❖ **Future Scope:**

- AI-based arrival prediction
- Mobile app with notifications
- Voice announcements at bus stops
- Integration with Google Maps